

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Define Problem Statements

ShopEZ — MERN Stack E-commerce Application

Phase: Brainstorming & Ideation Phase

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Define Problem Statements

1. Introduction

This document formalises the problem statements identified during the Brainstorming & Ideation phase. Each statement follows the structured "**I am / I'm trying to / But / Because / Which makes me feel**" framework to ensure user-centric problem articulation. Root-cause analysis is subsequently applied to identify systemic factors driving these problems.

2. Problem Statements

PS-1: The Online Shopper

Field	Description
I am	A busy online shopper seeking electronics and clothing items
I'm trying to	Find and purchase specific items quickly and securely from a trusted platform
But	Existing stores suffer from poor search functionality, confusing multi-step checkouts, and loss of cart state across sessions
Because	Their systems are inadequately optimised — they lack advanced filtering, cross-device cart persistence, and seamless payment gateway integration
Which makes me feel	Frustrated, distrustful of the platform, and highly likely to abandon my cart prior to purchase

Impact Severity: ● High — Cart abandonment directly translates to revenue loss for the business.

PS-2: The Store Administrator

Field	Description
I am	A business owner and store administrator responsible for daily operations
I'm trying to	Manage product inventory efficiently, track monthly sales, and ensure timely order fulfilment
But	The process is excessively time-consuming — updating product details, uploading images, and accessing sales insights requires navigating a clunky, non-intuitive dashboard
Because	Current systems lack a unified admin interface; image uploads are not CDN-optimised, and sales data is not aggregated in a readily interpretable format
Which makes me feel	Overwhelmed, inefficient, and deeply concerned about the scalability of business operations

Impact Severity: ● High — Operational inefficiency leads to delayed inventory updates, missed sales opportunities, and reduced business agility.

3. Root-Cause Analysis (Ishikawa Diagram)



FIGURE 2.1 — ISHIKAWA (FISHBONE) ROOT-CAUSE DIAGRAM

4. Problem-to-Feature Mapping

Root Cause	Proposed Solution Feature
No SPA / Slow page reloads	React.js Single Page Application with Virtual DOM
No persistent cart state	MongoDB-backed cart schema linked to authenticated user session
Poor search & filter	Full-text search with price, category, and rating filters
Confusing checkout	Guided 3-step checkout wizard (Address → Method → Payment)
No CDN for images	Cloudinary integration for optimised image delivery
No trusted payment gateway	Stripe / Razorpay payment gateway integration
Unencrypted passwords	Bcrypt password hashing before persistence
No JWT auth	JWT tokens stored in HTTP-only cookies for XSS prevention
No admin dashboard	Dedicated /admin panel with sales analytics and order management

5. Conclusion

The two problem statements — representing the **end consumer** and the **store operator** — collectively define the dual-focus of the ShopEZ platform. All identified root causes have corresponding solution features mapped in the *Proposed Solution* and *Requirement Analysis* documents.

